

Department of Economics and Business Economics
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Research Interests

Network analysis, time series, econometrics, statistics, forecasting, machine learning

Academic Positions and Affiliations

Research Fellow	2025–
<i>Aarhus Center for Econometrics (ACE), Aarhus University</i>	
Associate Professor	2022–
<i>Department of Economics and Business Economics, Aarhus University</i>	
Senior Member	2022–2025
<i>DNRF Chair in Econometrics, Aarhus University</i>	
Assistant Professor	2018–2022
<i>Department of Economics and Business Economics, Aarhus University</i>	

Education

Ph.D. in Economics (<i>cum laude</i>)	2014–2018
Thesis Title: Essays in Network Modelling	
Supervisors: Christian Brownlees and Gabor Lugosi	
<i>Department of Economics and Business, Universitat Pompeu Fabra</i>	
M.Sc. in Macroeconomic Policy and Financial Markets	2012–2013
<i>Barcelona Graduate School of Economics</i>	
M.Sc. in Economics	2010–2011
<i>London School of Economics</i>	
B.Sc. in Economics	2007–2010
<i>University of Iceland</i>	

Publications

1. **Guðmundsson, G. S.** (2026). Detecting Giver and Receiver Spillover Groups in Large Vector Autoregressions. *Journal of Business and Economics Statistics*, 44:1, 297–308. DOI: [10.1080/07350015.2025.2526430](https://doi.org/10.1080/07350015.2025.2526430)
2. Brownlees, C. & **Guðmundsson, G. S.** (2025). Performance of Empirical Risk Minimization for Linear Regression with Dependent Data. *Econometric Theory*, 41:2, 391–420. DOI: [10.1017/S0266466623000348](https://doi.org/10.1017/S0266466623000348)
3. Brownlees, C., **Guðmundsson, G. S.** & Lugosi, G. (2022). Community Detection in Partial Correlation Network Models. *Journal of Business and Economics Statistics*, 40:1, 216–226. DOI: [10.1080/07350015.2020.1798241](https://doi.org/10.1080/07350015.2020.1798241)

4. **Guðmundsson, G. S.** & Brownlees, C. (2021). Detecting Groups in Large Vector Autoregressions. *Journal of Econometrics*, 225:1, 2–26. DOI: [10.1016/j.jeconom.2021.03.012](https://doi.org/10.1016/j.jeconom.2021.03.012)
5. **Guðmundsson, G. S.** & Zoega, G. (2016). A Double-Edged Sword: High Interest Rates in Capital Control Regimes. *Economics*, 10, 1–38.
6. **Guðmundsson, G. S.** & Zoega, G. (2014). Age Structure and the Current Account. *Economics Letters*, 123, 183–186.

Working Papers

- Performance of Empirical Risk Minimization for Principal Component Regression
with Christian Brownlees and Yaping Wang
- Community Detection in High-Dimensional Non-Stationary Time Series
with Majid M. Al-Sadoon
- Concentration of the Adjacency and Laplacian Matrices in Random Graphs with Dependent Edges
with Christian Brownlees
- k -Means Asymptotics
with Vanessa Berenguer-Rico and Bent Nielsen

Seminar and Conference Presentations

- Seminar at the Toulouse School of Economics, 26th November, 2024.
- Presentation at the 14th Workshop in Time Series Econometrics, University of Zaragoza, 11-12th April, 2024.
- Presentation at the 17th International Conference on Computational and Financial Econometrics, HTW Berlin, University of Applied Sciences, 16-18th December, 2023.
- Presentation at the 2nd Academic Workshop for Icelandic Economists Abroad, University of Iceland, 5th May, 2023.
- Seminar at the Department of Data Science and Analytics at BI Norwegian Business School, 18th April, 2023.
- Presentation at the 3rd High Voltage Econometrics conference, 7-8th October, 2022.
- Invited talk at the Statistical Methods on Networks Workshop, University of Leipzig, 26–28th September, 2022.
- Presentation at the 12th Workshop in Time Series Econometrics, University of Zaragoza, 31st March–1st April, 2022.

Seminar at the Department of Economics at Lund University, (27th September, 2019); Poster at the 2019 NBER-NSF Time Series Conference, Chinese University of Hong Kong (14–15th August, 2019); Presentation at the 12th Annual Meeting of the Society for Financial Econometrics (SoFiE), Fudan University (12–14th June, 2019); Presentation at the Academic Workshop for Icelandic Economists Abroad, University of Iceland, (27th May, 2019); Presentation at the 9th Workshop in Time Series Econometrics, University of Zaragoza (4–5th April, 2019); Presentation at the 12th International Conference on Computational and Financial Econometrics, University of Pisa (14–16th December, 2018); Presentation at the 8th Workshop in Time Series Econometrics, University of Zaragoza (12–13th April, 2018); Presentation at the 11th International Conference on Computational and Financial Econometrics, University of London (16–18th December, 2017);

Presentation at the 3rd Vienna Workshop on High-Dimensional Time Series in Macroeconomics and Finance, Institute for Advanced Studies Vienna (8–9th June, 2017).

Conference and Workshop Organisation

Organisation

- 1st Workshop of the Icelandic Econometrics Society, Reykjavík, June 20th, 2025.
- 4th Aarhus Workshop in Econometrics (AWE), Aarhus, October 11th, 2024.

Refereeing

Biometrika, International Journal of Forecasting, Journal of Econometric Methods, Journal of Econometrics, Statistical Papers.

Teaching

Graduate

4616: Time Series Econometrics	Aarhus University	2023–
4645: Machine Learning Methods in Empirical Economics	Aarhus University	2022–
Applied Data Science (Network Analysis)	Aarhus University	2020–2022
4616: Time Series Econometrics	Aarhus University	2018–2019

Undergraduate

3620: Econometrics I	Aarhus University	2025
Introduction to Machine Learning	Aarhus University	2020–
3620: Econometrics I	Aarhus University	2018–2021

Awards and Scholarships

- Nominated for Den Gyldne Pegepind (teacher of the year) at the Department of Economics, Aarhus University, 2020.
- Nominated for Den Gyldne Pegepind (teacher of the year) at the Department of Economics, Aarhus University, 2019.
- Marcelo Reyes Prize for best paper by junior researcher at the 8th Workshop in Time Series Econometrics, the University of Zaragoza, April 2018.
- FPU Scholarship, the Spanish Ministry of Education, Culture and Sports, 2015–2018.

Other

- Bachelor's, Master's and Ph.D. Supervision Programme, Aarhus University, 2024.
- University Pedagogical Programme, Aarhus University, 2021.
- Reviewer for Spring Meeting of Young Economists, 2018.
- Initial Training in University Teaching, Universitat Pompeu Fabra, 2017.